

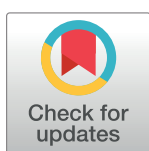
RESEARCH ARTICLE

Experimental study on the properties of ultra-high-strength geopolymer concrete with polypropylene fibers and nano-silica

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Abstract

Because of the recent progress in materials properties, specifically high-strength concrete, further research is needed to evaluate its suitability, understanding, and performance in the modern-day world. This research aims to enhance the performance of ultra-high-strength geopolymer concrete (UHS-GPC) by adding nano-silica (NS) and polypropylene fibers (PPFs). Three 1%, 2%, and 3% different amounts of PPFs and three NS 5%, 10%, and 15% were utilized in the samples. Various performance parameters of UHS-GPC were evaluated, such as fresh property, compressive strength, modulus of elasticity split tensile, flexural and bonding strength, drying shrinkage, load-displacement test, fracture performance, and elevated temperature. The test outcomes showed that by raising the percentage of PPFs and NS to the allowable limit, the performance of UHS-GPC can be improved significantly. The most improved performance of UHS-GPC was obtained at 2% polypropylene fibers and 10% nano-silica, as the compressive, splitting tensile, flexural. Bond strength was improved by 17.07%, 47.1%, 36.52, and 37.58%, and the modulus of elasticity increased by 31.4% at 56 days. The study showed that the sample with 2% PPFs and 10% NS had excellent performance in the load-displacement test, drying shrinkage, fracture behavior, and elevated temperature. At 750°C elevated temperature, the samples' strength was reduced drastically, but at 250°C, the modified samples showed good resistance to heat by retaining their compressive strength to some degree. The present work showed the suitability of PPFs and NS to develop ultra-high-strength geopolymer concrete, which can be used as a possible alternate material for Portland cement-based concrete.

1. Introduction

Ultra-high-strength concrete is suitable for constructing longer bridges, tall buildings, and structures exposed to aggressive conditions [1]. However, the mass of ordinary Portland

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cement in ultra-high-strength concrete (UHSC) is usually 800 to 1150 kg/m³ [2], which is twice or thrice the proportion of conventional cement in traditional concrete [3–5]; the formation of OPC needs a considerable proportion of energy and natural resources [6], which makes the immense proportion of CO₂ [7]. Making 1 ton of clinker is projected to need 6.5 to 7 mega-joules of energy and releases nearly [8] 0.90 to 1.0 tons of CO₂. The engineers tried to reduce the volume of binders and replace additional binder-based materials with ordinary Portland cement [9–11]. Yu et al. [12] revealed that HSC was developed with a lowered binder volume of 680 kg/m³, which led to a 29% decrease in the release of CO₂ to the outer atmosphere [13]. Compared to ordinary Portland cement, geopolymer is a binder that is mostly clinker free and low carbon-based [14]. It can be developed by chemical activation of materials that are surplus in aluminosilicates, for instance, granulated blast furnace slag (GBFS) [15], meta halloysite [16], wheat straw ash [17], metakaolin (MK) [18, 19] silica fume (SF) [20], and these materials can be activated by utilizing solutions such as sulfate, sodium hydroxide, silicates, etc. Laterite soil can also be a potential material for developing geopolymer material strength properties similar to cement-based concrete [21]. GPC has received massive consideration because of the considerable decrease in the proportion of carbon dioxide outflow and the requirement for natural raw materials [22–25]. Contrasting to Portland cement, the manufacturing of raw materials doesn't need a calcination method which leads to a decrease in energy consumption [26, 27]. It is observed from past studies that geopolymer concrete releases five to six times less carbon dioxide than conventional concrete. The current progress in the geopolymer concrete domain demands ultra-high-strength geopolymer concrete (UHS-GPC) by utilizing geopolymer (GP) as a binding material [28, 29]. Authors in a study [25] displayed that the highest bending and compression strength of UHS-GPC with 3% fibers were 13 MPa and 181 MPa after curing for 28 days, correspondingly when it was chemically activated with solutions of sodium hydroxide and sodium silicate. Toniolo [30] and Ahmed [31] also used nano-silica and slag to develop UHS-GPC, attaining compression strength of greater than 145 MPa. Several research studies have shown that only slag-based geopolymers lead to complications such as reduced workability, high shrinkage, fast setting time [32], and loss of strength properties during carbonation. Mixing slag and silica fume seem very appropriate compared to employing only slag for attaining excellent durability, strength, and new properties of GPC [33]. There appears to be significant room for enhancement in these domains of UHS-GPC.

Mineral admixtures or fillers are supplementary cementitious materials (SCMs) [34] and are utilized to reduce the cost, improve the slump value, and harden the characteristics of concrete [35–37]. Wheat straw ash (WSA), silica fume (SF), rice husk ash (RHA), fly-ash (FA) and granulated blast furnace slag (GBFS) have been utilized in past research extensively as SCMs because of their positive effect on the strength, durability characteristics, cost, and environment. The two significant elements of UHS-GPC are nano-silica (NS) and polypropylene fibers (PPFs). The inclusion of PPFs enhances ductility and resistance against impact considerably. Zaid et al. [14] indicated that when the fibers by 4% weight of binder were utilized in geopolymer composites, it had an irrelevant impact on the strength and ductility properties of geopolymer concrete. Polypropylene fibers (PPFs) improve concrete's toughness, ductility, and post-cracking behavior [38]. The optimizing and design method to attain the optimal performance of UHS-GPC with fibers depends on the physical properties of fibers. Furthermore, the comparatively higher price of fibers and appropriate control of their mixing in a GPC matrix is essential for real-life usage and commercialization of UHS-GPC. NS plays a crucial role in HSC throughout its packed densifying effect. Furthermore, research indicated that including 10 to 15% NS in ultra-high-strength concrete enhances the freshness of concrete. Some past research has shown that introducing NS to GPC improved strength formation. However, it reduced the slump value [24, 39]. Das et al. [20] showed that introducing an

appropriate proportion of NS enhanced the strength and fresh characteristics of UHS-GPC. In cement-based ultra-high-strength concrete, nano-silica mainly enhances strength and durability due to its nano-size and densifying influence [40, 41]. A past study observed enhancement in fresh properties of ultra-high-strength concrete with nano-silica [42]. For GPC, though different past studies have revealed that the introduction of nano-silica helped in the strength gain, the rheological properties were reduced [43–46]. Nematzadah et al. [47] showed that adding an optimal dose of NS can enhance geopolymer concrete's strength and fresh properties. However, Chithra et al. [48] observed that substituting fly ash with nano-silica in concrete led to a reduction in fresh characteristics of GPC. The utilization of nano-silica in fly-ash-based GPC was investigated in some of the past studies, though because of the lack of knowledge on research related to the strength, ductility, and fracture characteristics of GBFS/SF based GPC modified with nano-silica and PPFs are very limited in the literature, GPC is not utilized in the construction sector and building codes. Drying shrinkage considerably impacts the durability characteristics of concrete [49]. The concrete drops water by the capillary action during drying, which causes cracks and shrinkage [50–53]. Authors in a study [54] revealed a considerable decrease in shrinkage when PPFs or steel fibers were employed as strengthening materials in concrete. The research concluded that adding more proportion of fibers up to a certain amount reduces drying shrinkage.

1.1 Novelty of present work

Further research is necessary because of the advancement and challenges related to the GPC, particularly the different properties of ultra-high-strength fiber-reinforced geopolymer concrete (UHS-GPC). This research aims to elevate the awareness of UHS-GPC by examining the influence of nano-silica (NS) and polypropylene fibers (PPFs) on ultra-high-strength geopolymer concrete's strength and fracture characteristics. PPFs were added in 1%, 2%, and 3% by the binder's weight, and NS was added in 5%, 10%, and 15% by the binder's weight. As per the author's reliable information, no considerable research has been done on the effect of nano-silica on the properties of ultra-high-strength geopolymer concrete modified with polypropylene fibers by evaluating the flow diameter, compression strength, modulus of elasticity, split tensile strength and flexural strength, pullout test, load deflection capacity, drying shrinkage, fracture energy, and lastly the elevated temperature of ultra-high-strength fiber-reinforced geopolymer concrete, which shows the originality of the current work. The present research will assist in growing information on developing ultra-high-strength fiber-reinforced geopolymer concrete with excellent properties in different strength characteristics and against elevated temperatures.

2. Raw materials and sample properties

Table 1 presents the physical and Table 2 shows the chemical properties of silica fume, nano-silica, and granulated blast furnace slag. The chemical composition of silica fume, nano-silica, and GBFS is obtained from x-ray diffraction (XRD) analysis as presented in Fig 1(A)–1(C). The images of silica fume, nano-silica, and GBFS are provided in Fig 2(A)–2(C). Na_2SiO_3 and NaOH were utilized as alkali chemicals. The particle size distribution for the fine and coarse aggregate is presented in Fig 3(A) and 3(B). To start, SF, NS, and GBFS were dry mixed for five minutes, and then river sand (fine aggregate) was added and further dry mixed for three minutes. The alkaline solutions were then added and mixed for three minutes. The Na_2SiO_3 and NaOH were obtained from the local chemical industry in Islamabad. Polypropylene fibers (PPFs), as presented in Fig 2(D), were used to improve the ductility of geopolymer concrete. The physical characteristics of PPFs are displayed in Table 3. Fibers were added in 1%, 2%, and

Table 1. Physical characteristics of nano-silica, silica fume, GBFS.

Physical Property	Nano-silica	Silica Fume	GBFS
Mean particle size (microns)	0.20	37	16
Surface area (m ² /kg)	1858	448	324

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3%, and nano-silica was introduced in 5%, 10%, and 15%. Complete mix details of all the samples are displayed in Table 4. The sample's terminology was designed to have four groups in total. The first is the control samples used for reference, and then for modified samples, the term "G" denotes the group. The number after the "G" indicates the group number. Similarly, the word "PPFs" and "NS" means polypropylene fibers and nano-silica in each group, and the number after the PPFs and NS indicates the percentage of these materials. All the samples were formed at ambient temperature.

3. Test procedures

Numerous tests were performed in the present research to evaluate the fresh and strength properties of polypropylene fibers-reinforced UHS-GPC. The following segments highlight the test methods.

3.1 Fresh property

The influence of adding fiber on the fresh properties of ultra-high-strength GPC was evaluated as flow diameter following BS EN 1015–11 [55]. After mixing the mixtures, it was poured into a cone and lifted and spread onto the flow table. The diameter of the pat on two sides at 90 angles to each other was evaluated. The flow test was performed when mixing was completed for every batch of samples, and every mix of samples was tested three times to bring uniformity to the testing.

3.2 Compression strength performance and Modulus of Elasticity (MOE)

This research determined the compression performance of entire specimens by utilizing MOE and compression strength. Cylinders of 6 x 12 inches (length x diameter) were used to determine the concrete's compressive strength following ASTM C 39 [56]. To evaluate MOE, 6-inch x 12-inch concrete cylinders were used following ASTM C 469 [57]. The rate of loading for compression and MOE was 1.5 MPa/sec. To obtain MOE, a steel ring with a strain gauge was fitted around the concrete cylinder, and the stress-strain data of specimens were obtained.

Table 2. Chemical properties of silica fume, GBFS, nano-silica.

Chemical elements	Silica Fume	GBFS	Nano-Silica
Silica (SiO ₂)	68.87	31.47	95.25
Al ₂ O ₃	20.46	15.62	-
MgO	0.51	6.72	0.25
Fe ₂ O ₃	2.61	0.54	0.49
CaO	1.78	39.64	1.62
K ₂ O	2.01	0.71	0.69
SO ₃	0.39	2.62	-
Na ₂ O	1.48	0.29	0.64
Loss of ignition	1.89	2.39	1.06

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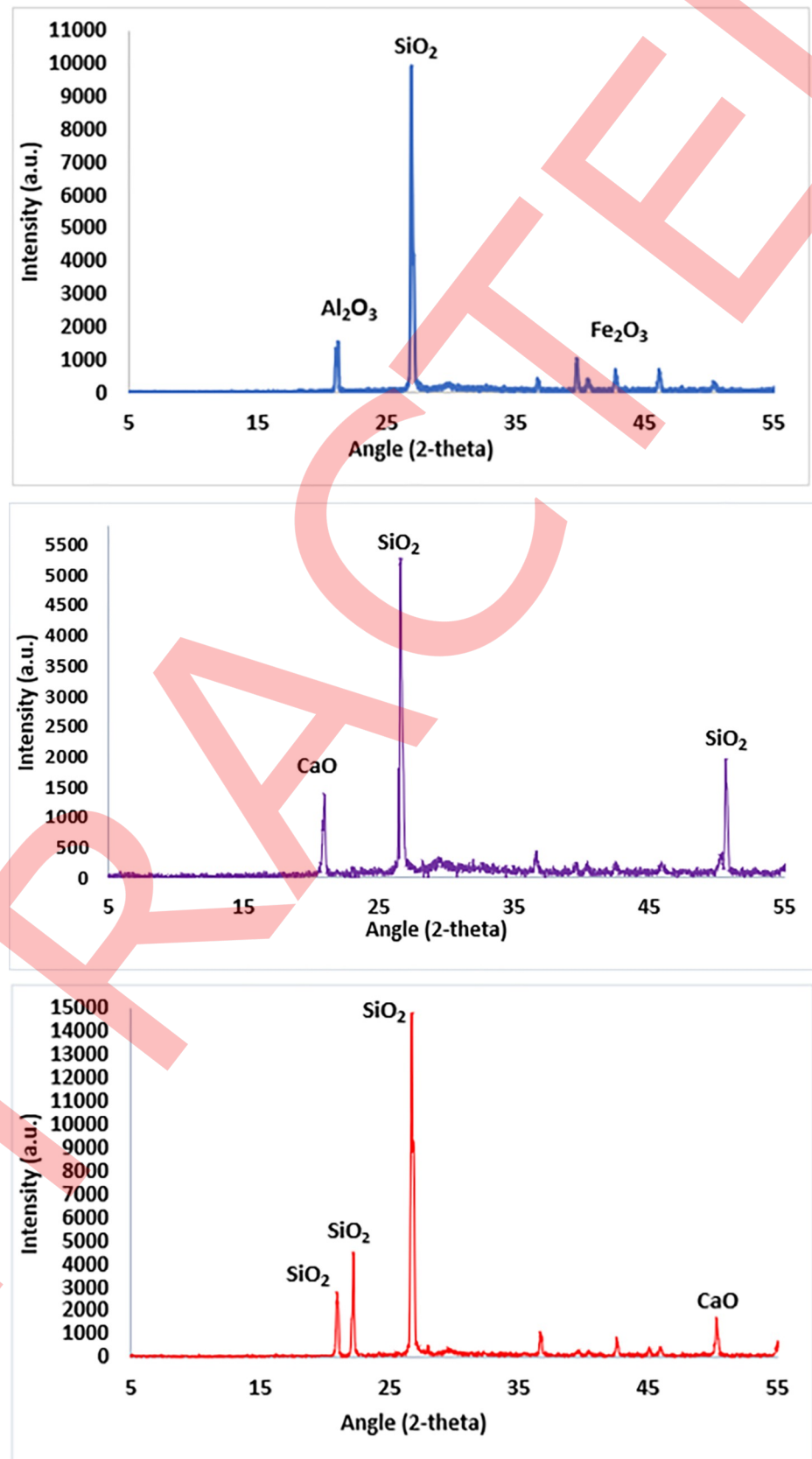


Fig 1. XRD analysis of (a) Silica Fume (SF), (b) Granulated blast furnace slag (GBFS), and (c) Nano-silica (NS).

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Fig 2. Images of (a) Silica Fume (SF), (b) Nano-Silica (NS), (c) Granulated Blast Furnace Slag (GBFS), and (d) Polypropylene Fibers (PPFs).

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Compressive strength of HS-GPC was evaluated at 28, 56, and 90 days of age, and MOE was assessed at 28 and 56 days of age.

3.3 Pullout test

To determine the bond strength of UHS-GPC, a pullout test was carried out following ASTM C 900 [58]. Concrete cubes of 6 x 6 x 6 inches were arranged and tried at the curing of 28 and 56 days. To achieve a smooth sample surface, the top layer of the concrete specimen was capped with gypsum to distribute a uniform load. The test setup for the pullout test is presented in Fig 4. The bond strength of concrete specimens was evaluated under the pullout test using the following Formula (A).

$$\tau = \frac{F}{3.14 * L * d} \quad (A)$$

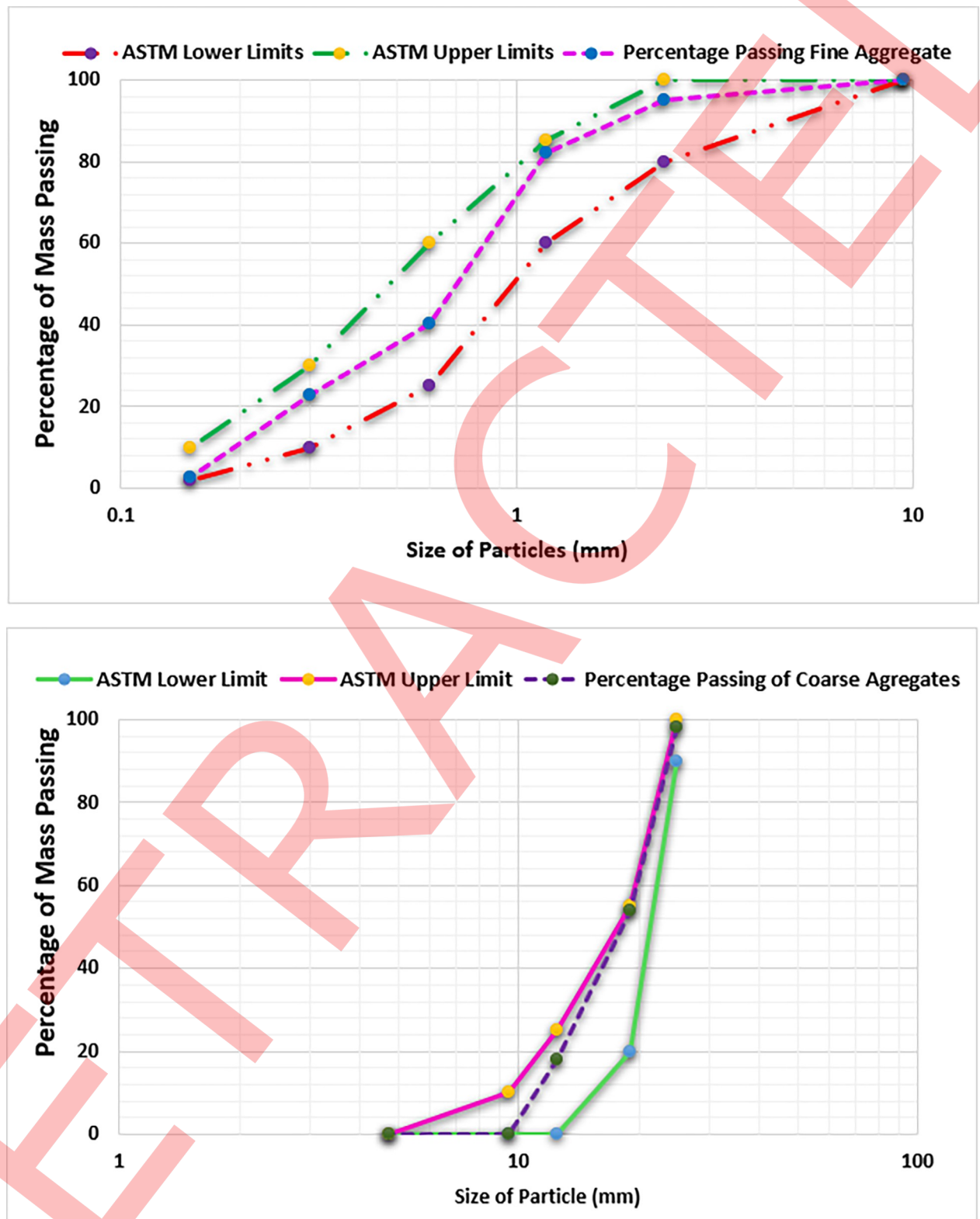


Fig 3. Particle Size Distribution; (a) Fine aggregates, (b) Coarse aggregates.

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Table 3. Physical characteristics of polypropylene fibers (PPFs).

Physical Characteristics	Result
Color	White
Length (mm)	20
Tensile Strength (MPa)	600 to 650
Young's Modulus (GPa)	4.0
Density (g/cm ³)	1.1

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Here, L is the embedded length of steel reinforcement, F is the force, and d is the steel diameter in mm.

3.4 Splitting tensile strength performance

To evaluate the splitting tensile strength (STS) of all the samples, a universal testing machine (UTM) with a limit of 1200 kN. was employed following ASTM C 496 [59]. The rate of loading from splitting tensile strength was 0.15 MPa/sec. Concrete samples of 6-inch x 12-inch were used to find the STS. Splitting tensile strength was evaluated at 28 days of strength. The splitting tensile strength is evaluated using the following relationship (B).

$$f_{st} = 0.63 \frac{F}{A} \quad (B)$$

In the above relationship, A is the concrete's x-section area in mm², F is the cracking load in Newton, and f_{st} is splitting tensile strength.

3.5 Flexural behavior

To assess the flexural strength of UHS-GPC, 6 x 6 x 24 inches concrete beams were prepared to perform the three-point bending test following ASTM C78 [60]. A universal testing machine was also used for this with the assembly of the flexural tests. A linear variable differential transducer (LVDT) was installed at the mid-position of the beam soffit. This test's highest loading rate was 0.55 mm/minute, with the highest force of 190 kN. The test setup for flexural is presented in Fig 5. The flexural strength was evaluated at curing of 28 days of strength. The

Table 4. Mix design of materials (kg/m³).

Mixing ID	SF	GBFS	NS	SS	SH	PPFs	Fine Aggregate	Coarse Aggregate	Water
Control	165	685	0	65	130	0	455	650	175
G2-PPFs-1-NS-5	165	685	42.5	65	130	8.5	455	650	175
G2-PPFs-2-NS-5	165	685	42.5	65	130	17	455	650	175
G2-PPFs-3-NS-5	165	685	42.5	65	130	25.5	455	650	175
G3-PPFs-1-NS-10	165	685	85	65	130	8.5	455	650	175
G3-PPFs-2-NS-10	165	685	85	65	130	17	455	650	175
G3-PPFs-3-NS-10	165	685	85	65	130	25.5	455	650	175
G4-PPFs-1-NS-15	165	685	127.5	65	130	8.5	455	650	175
G4-PPFs-2-NS-15	165	685	127.5	65	130	17	455	650	175
G4-PPFs-3-NS-15	165	685	127.5	65	130	25.5	455	650	175

SF–Silica Fume, GBFS–Granulated Blast Furnace Slag, SS–Sodium Silicate, SH–Sodium Hydroxide, PPFs–Polypropylene Fibers

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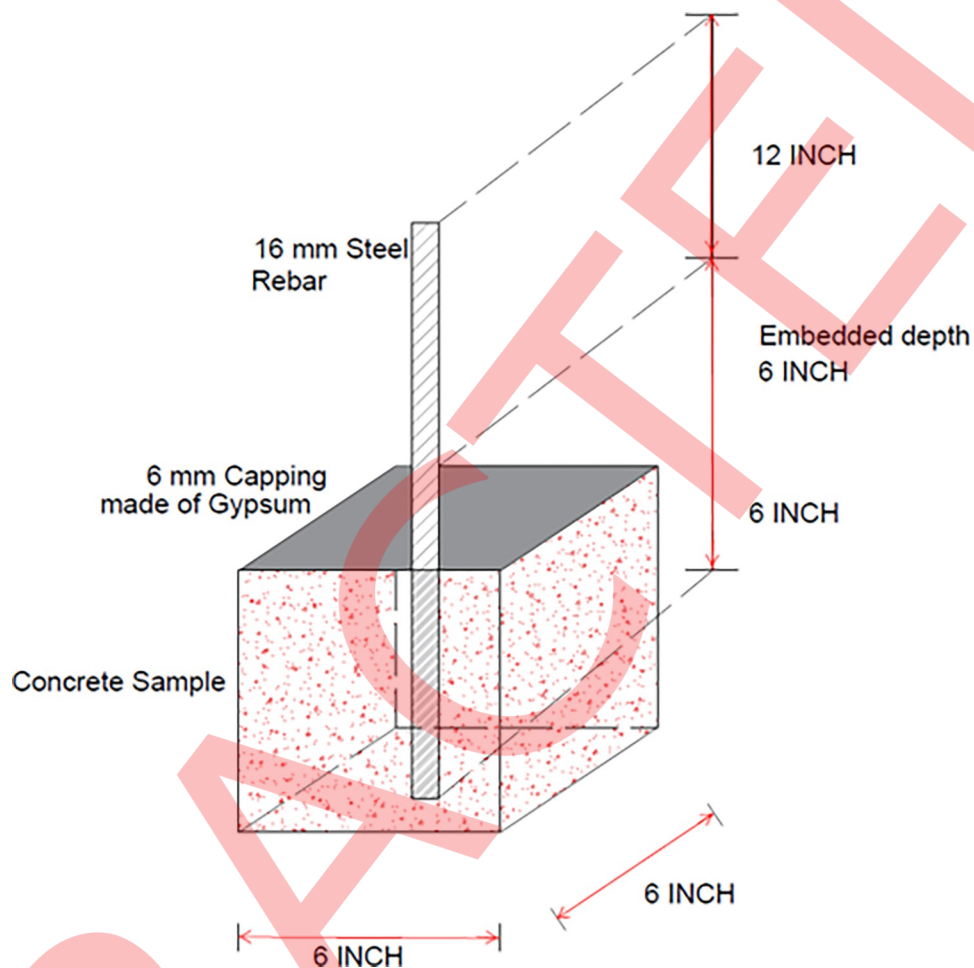


Fig 4. Test setup of bond strength.

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flexural strength of UHS-GPC is evaluated using the following relationship (C).

$$F_u = \frac{3 * fL}{(h - a)^2 * 2b} \quad (C)$$

In the above relationship, f is the ultimate highest load in Newton, F_u is the flexural strength, h is the sample's height, L is the clear distance, and a is the notch's height in the 3-point bending test.

To evaluate the ductility of UHS-GPC, 6 x 6 x 24 inches concrete beams were cast per ASTM C1609 [61]. A stiff plastic mold was prepared to confine any movements during the placement of the fresh mix. Before concrete casting, electrical strain gauges were installed with a gauge resistance of $119 \pm 0.5 \Omega$, and the molds were lubricated properly to de-mold the samples conveniently.

3.6 Drying shrinkage

This test is carried out to examine the change in the length of hardened concrete. Drying shrinkage was performed following ASTM C157 [62] and prismatic molds of 76 x 76 x 285

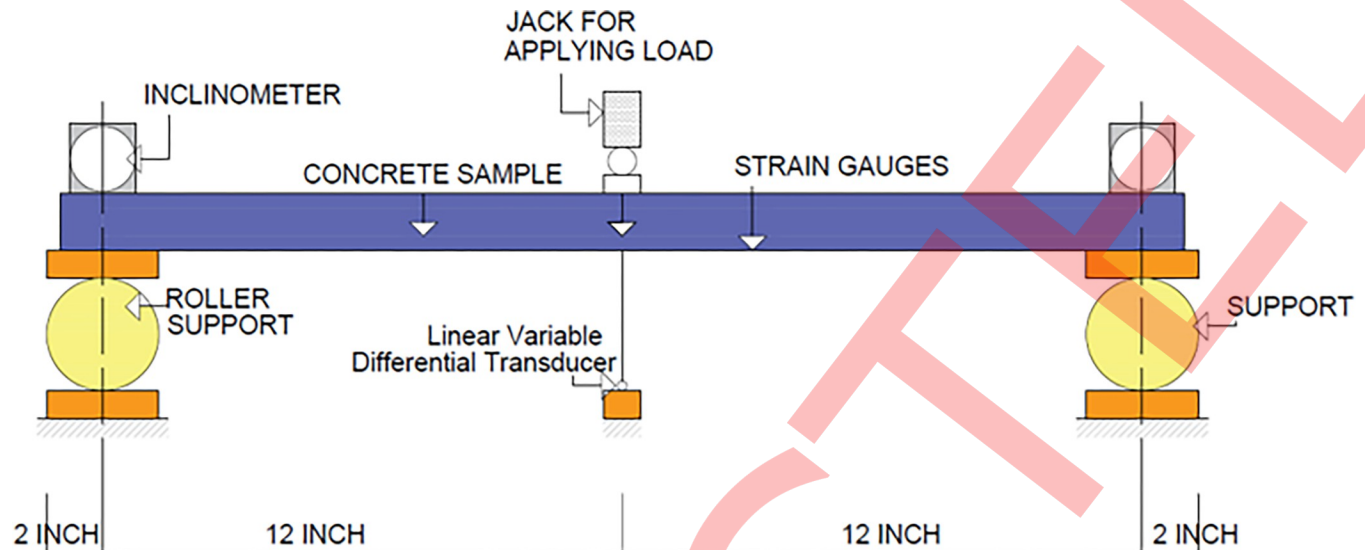


Fig 5. Test setup for flexural strength.

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mm. The test setup for drying shrinkage is presented in Fig 6. The samples were placed at a temperature of 24°C in a temperature-controlled cabinet and under moist conditions following ASTM C511 [63]. The drying shrinkage was assessed for 3, 7, 14, 21, 28, 56, and 90 days of age. The change in length was evaluated using the Formula (D) from ASTM C157 [62], and the results were depicted in the micro-strain.

$$\text{Change in length (micro-strain)} = \frac{(CRD_a - CRD_b)}{G} * 10^6 \quad (D)$$

In the above formula, G is the gauge length in mm, CRD_a is reading at a specific age in mm, and CRD_b is the reading after 24 hours of mixing in mm.

3.7 Fracture energy

In this test, a servo-controlled hydraulic jack is utilized on the concrete sample suspended on the beam. The beam was 400 mm in length with 100 mm in depth. Fracture energy was evaluated by installing a linear variable differential transducer (LVDT) on the beam subjected to a four-point bending test. Its test setup is presented in Fig 7. Fracture energy signifies the energy needed to develop a concrete matrix crack, which is evaluated by the suggested RILEM (E).

$$G_f = \frac{(W_o + mg\sigma_s)}{A} \quad (E)$$

In the above relationship, G_f is the fracture energy, σ_s is the displacement, mg is the concrete's weight, w_o is the area under the force vs displacement curve, and A is the area.

The relationship in (E) evaluated the critical intensity factor.

$$K = \frac{3P(max)*l}{2*b*d^2} * \sqrt{ao} * (1.9 - 3.06 + 14.48 A^2 + 25A^4) \quad (F)$$

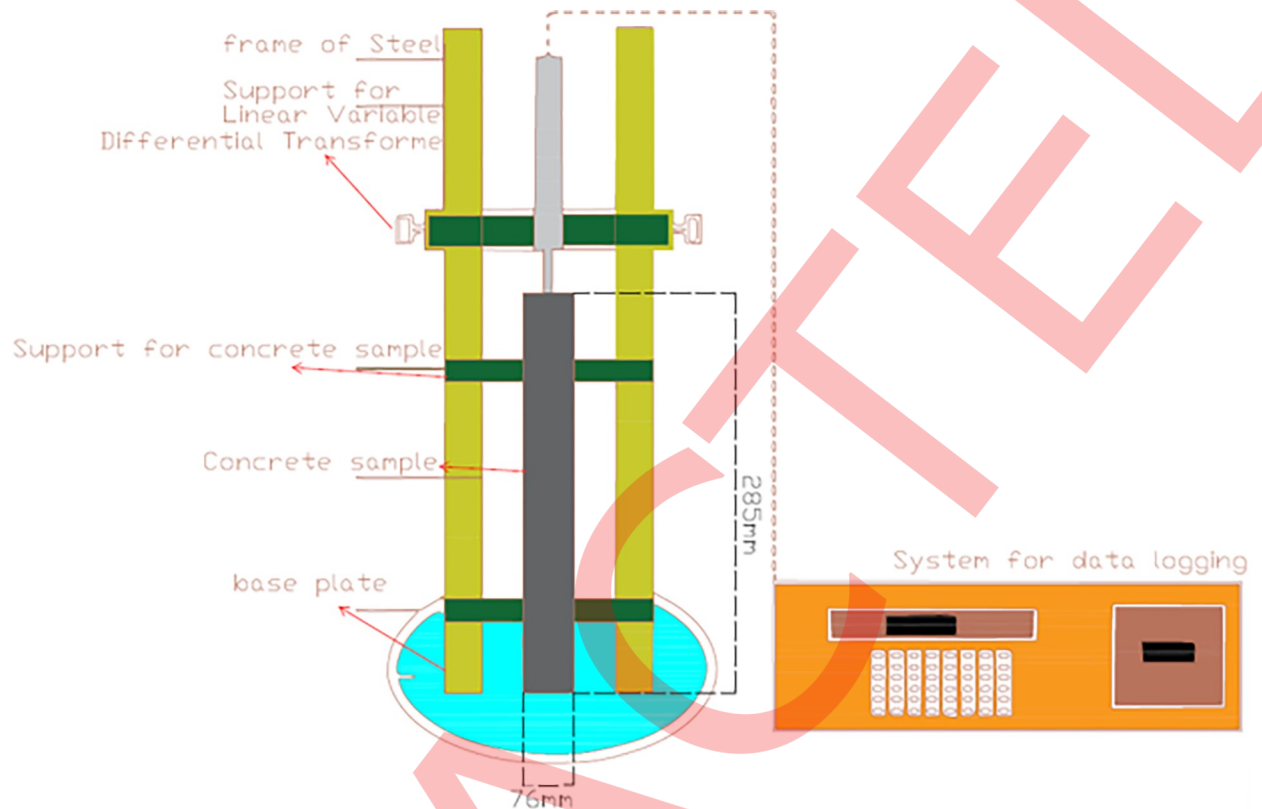


Fig 6. Test setup for drying shrinkage.

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In the above relationship (F), l is the sample's clear span, b is the sample's width, d is the sample's depth, P_{max} is the highest force, A is the ratio between the depth of notch and depth of the specimen, a_o is sample's depth of the notch.

3.8 Elevated temperature

In this test, concrete samples were exposed to high temperatures, and their effect on concrete compressive strength was evaluated. An electric plus gas furnace was employed for this test. When the desired heat was attained, the heat temperature was kept the same for 2.5 hours. The heating and cooling regime are presented in Fig 8. The concrete samples were exposed to 250°C, 500°C, and 750°C. The temperature was divided into three phases; the 1st phase shows the start of heating the temperature till it reached its desired temperature, 2nd phase shows the desired heat was sustained for 2.5 hours, and 3rd phase shows the cooling of the sample after heating it to the needed heat. Afterwards, the furnace was let cooled at the rate of 2.5°C per minute to evade thermal shock to the concrete samples. When the furnace was thoroughly cooled to ambient temperature, the samples were kept in the lab for 24 hours at ambient temperature. Then, the samples were tested to evaluate the loss in mass and strength of exposure to high temperatures.

4. Results and discussion

4.1 Fresh properties

The impact of adding PPFs on the fresh property of UHS-GPC is provided in Fig 9. It could be observed that the fresh characteristics of UHS-GPC changed significantly with the inclusion of

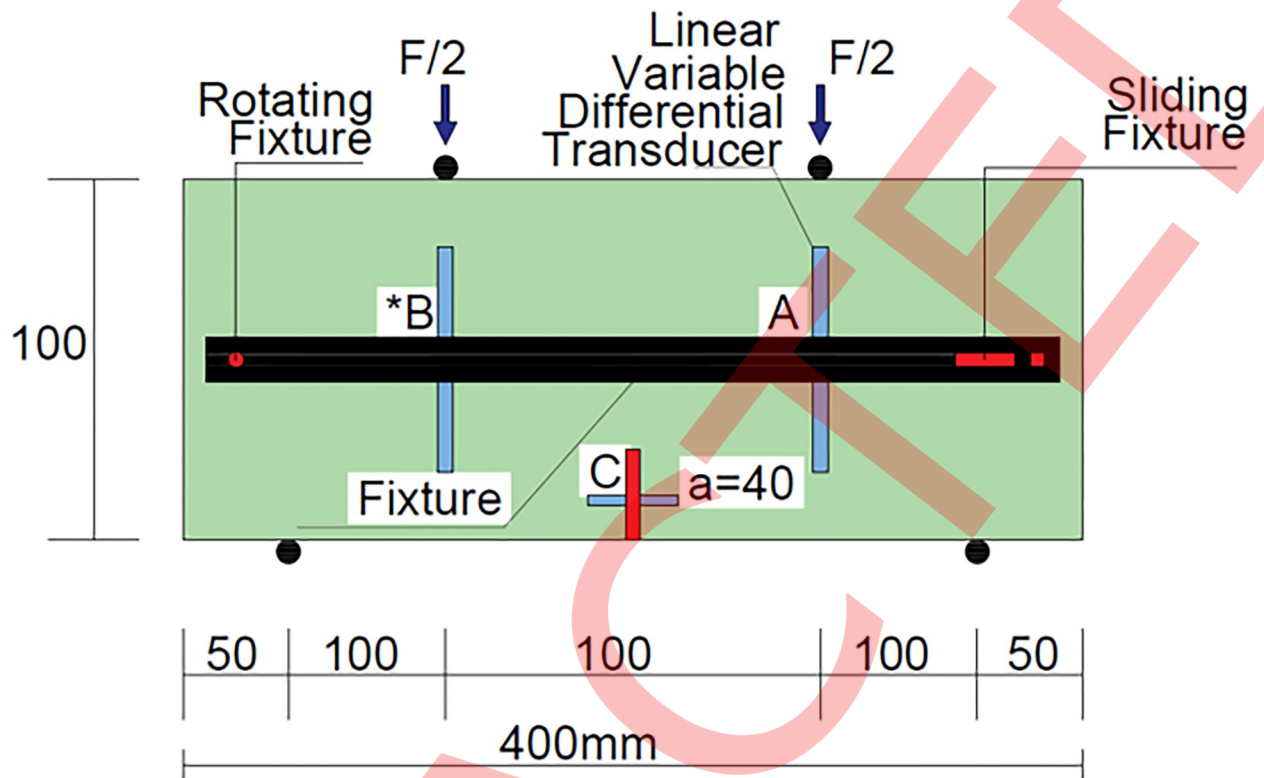


Fig 7. Test setup for fracture energy test.

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PPFs. It can be stated that the flow diameter was reduced when the proportion of fibers was raised. Furthermore, it should be kept in mind that the inclusion of a high percentage of PPFs, like 2% and 3% formed marginally dry mixtures in their fresh condition. Reduction in the flow with the incorporation of PPFs can be due to the high surface area of fibers which ingests the extra water available for lubricating the mixture. Ahmed et al. [64] revealed that increasing polypropylene fibers lead to a harsh mixture which can cause an issue in the strength or durability of hardened concrete. Changes in flow diameter were because of a rise in the amount of silica and its fast reactivity with the alkali solutions, where additional “SiO₂” behaved as nuclei for the precipitation of dissolved samples from silica fume the nano-silica. Das et al. [20] revealed that adding 10% lime as a substitute for fly ash to develop geopolymer concrete led to low workability. The decrease in the workability with adding the lime could be related to the increase in calcium in the concrete’s mix and the high dissolution of the monomers because of the rise in the alkalinity. The reduced value of concrete’s flowability could be due to raw materials’ physical and chemical properties [65]. Mustakim et al. [66] revealed that adding nano-silica after its optimal percentage will lead to lower workability; this could be because the nano-silica has a high surface area due to which the nano-silica absorbs extra accessible water from the mix, which makes the fresh concrete stiff and highly cohesive. Compared to granulated blast furnace slag, nano-silica has high reactivity, which could lead to fast agglomeration and result in concrete specimens with reduced values of flowability [67].

4.2 Compressive strength and modulus of elasticity

Fig 10(A) shows the co-relations between the modulus of elasticity (MOE) and compressive strength of UHS-GPC with different proportions of fibers. The MOE and compressive

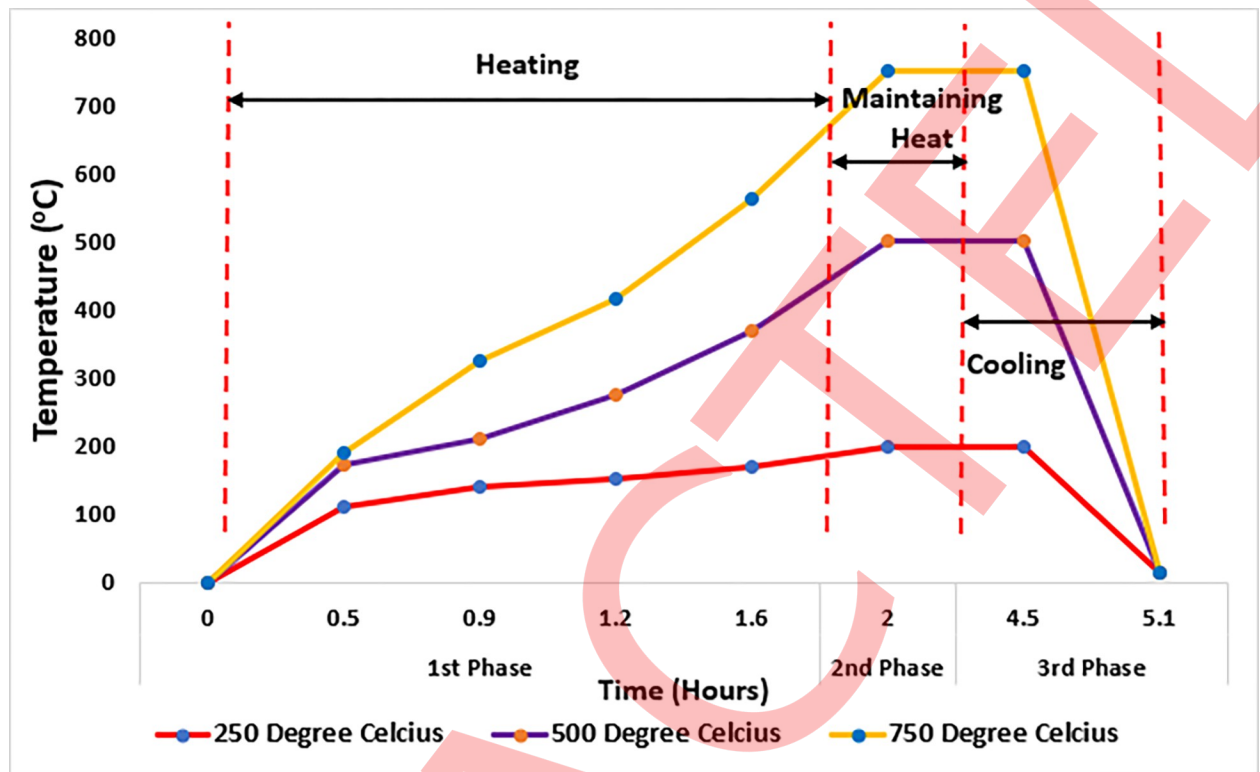


Fig 8. Heating and Cooling regime of elevated temperature.

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strength enhanced as the number of PPFs was raised. With no PPFs and nano-silica (control sample), the uppermost compressive strength at 90 days is 129.1 MPa. When the PPFs were 1%, the highest compressive strength at 90 days was 137.14 MPa which is a 5.86% improvement from than control sample. The compressive strength number was improved to 134.5 MPa, 146.4 MPa, and 155.4 MPa with the 2% PPFs and 10% nano-silica, and this was the highest compressive strength attained with 2% PPFs and 10% nano-silica, which is 16.72%, 17.07%, and 16.9% higher strength than samples with no PPFs at 28, 56 and 90 days. At 3% PPFs, the compressive strength of all the samples was lesser than those with a similar amount of nano-silica; adding more fibers causes the mix to be harsher by absorbing water which ultimately reduces the strength [68]. The tendency of strength in development in the samples was identical to previous research on fiber-reinforced GPC and ultra-high-strength concrete [69–71]. A higher amount of PPFs also decreases the space between fibers, which limits the onset and dispersion of cracks in the matrix. The effect of the addition of PPFs and nano-silica (NS) on the MOE of UHS-GPC samples is provided in Fig 10(B). The addition of NS led to a significant improvement in the modulus of elasticity. When 5% NS was utilized at 56 days, the highest modulus of elasticity was 30.7 GPa, but when 10% NS was added at 56 days, these strength values were enhanced by 14.1% and reached 35.75 GPa. But when the NS was increased from 10 to 15%, the modulus of elasticity reduced compared to other samples with a similar amount of PPFs. The samples with 10% NS and 2% PPFs had the highest modulus values of elasticity of 32 and 35.75 GPa at 28 and 56 days, which is more than 31.2% and 31.4% of the reference specimen at 28 and 56 days. Similar observations were reported in the literature [72, 73].

Co-relation analysis was performed between compressive strength and modulus of elasticity for 28 and 56 days, as presented in Fig 10(C). From the co-relation analysis, it can be observed

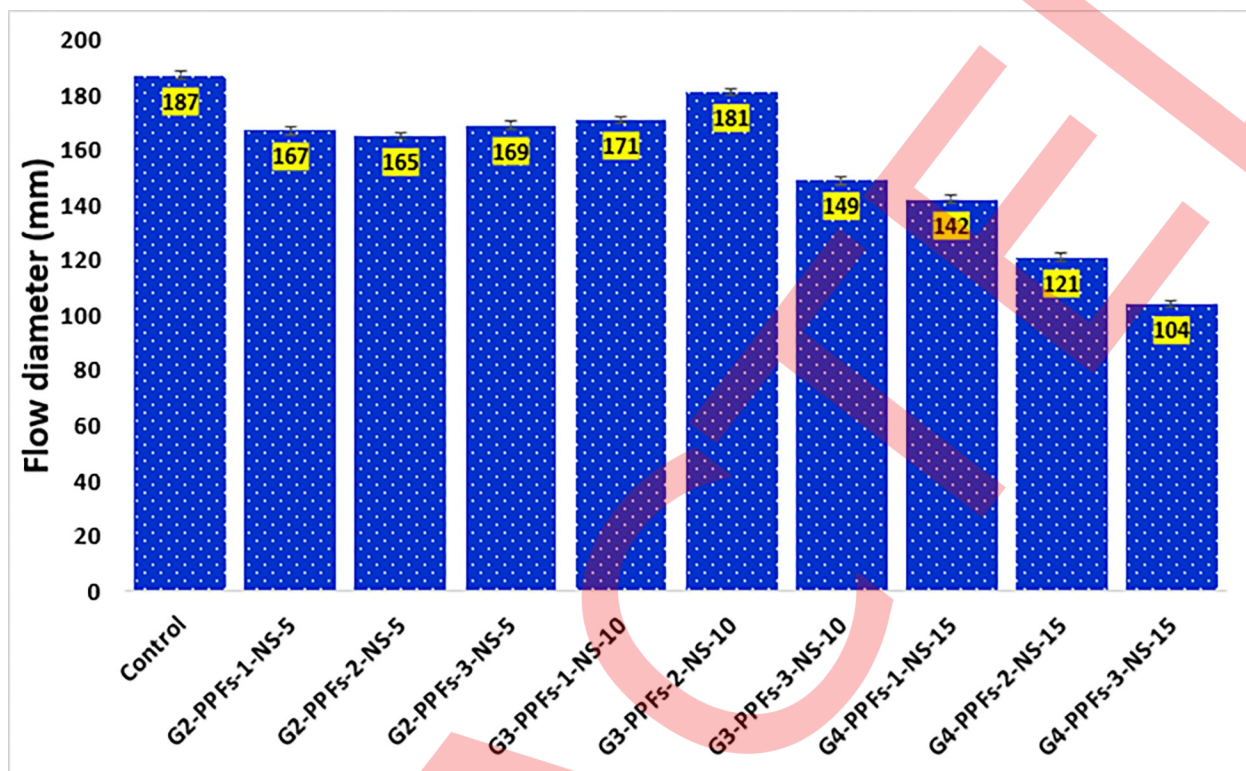


Fig 9. Impact of fibers and NS on flow diameter of UHS-GPC.

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that the relation between compressive strength and modulus of elasticity is very linear and shows the R^2 value of more than 75% for both the relationship, which shows the strong co-relation between the compressive strength and modulus of elasticity at 28 and 56 days.

4.3 Bonding strength

Fig 11 presents the changes in the bonding strength of samples because of the joint influence of nano-silica and polypropylene fibers. It could be said with confidence that bond strength improved considerably with adding polypropylene fibers. Furthermore, nano-silica also enhanced the UHS-GPC sample's bond strength. The maximum improvement of bond strength was observed to be 14.41 MPa and 17.3 MPa, which is the improvement of 26.47% and 37.58% at 28 and 56 days for the sample with 10% nano-silica and 2% PPFs in comparison with the reference sample. The effect of PPFs was noted to be more than nano-silica on the concrete's bond strength; this observation was also revealed in past studies. Ali et al. [74] showed that fibers improved the fiber's pullout strength by controlling the propagation of cracks in UHS-GPC specimens. The authors also revealed that the level of confinement at the fibers-concrete interface improved with adding fibers up to the optimal level, which enhances the concrete's bond strength and friction [27]. The authors in their study attained more than 25% of bond strength because of the inclusion of fibers. Several studies have examined the bond strength between steel rebars and conventional concrete [66, 67]. Still, there was no significant research on the bond strength between steel rebar and ultra-high-strength geopolymer concrete modified with nano-silica and PPFs. The present study will showcase the potency of nano-silica and PPFs on the bonding strength of UHS-GPC samples. In literature [27, 75], the authors revealed that the quality of concrete is a considerable parameter of the bonding

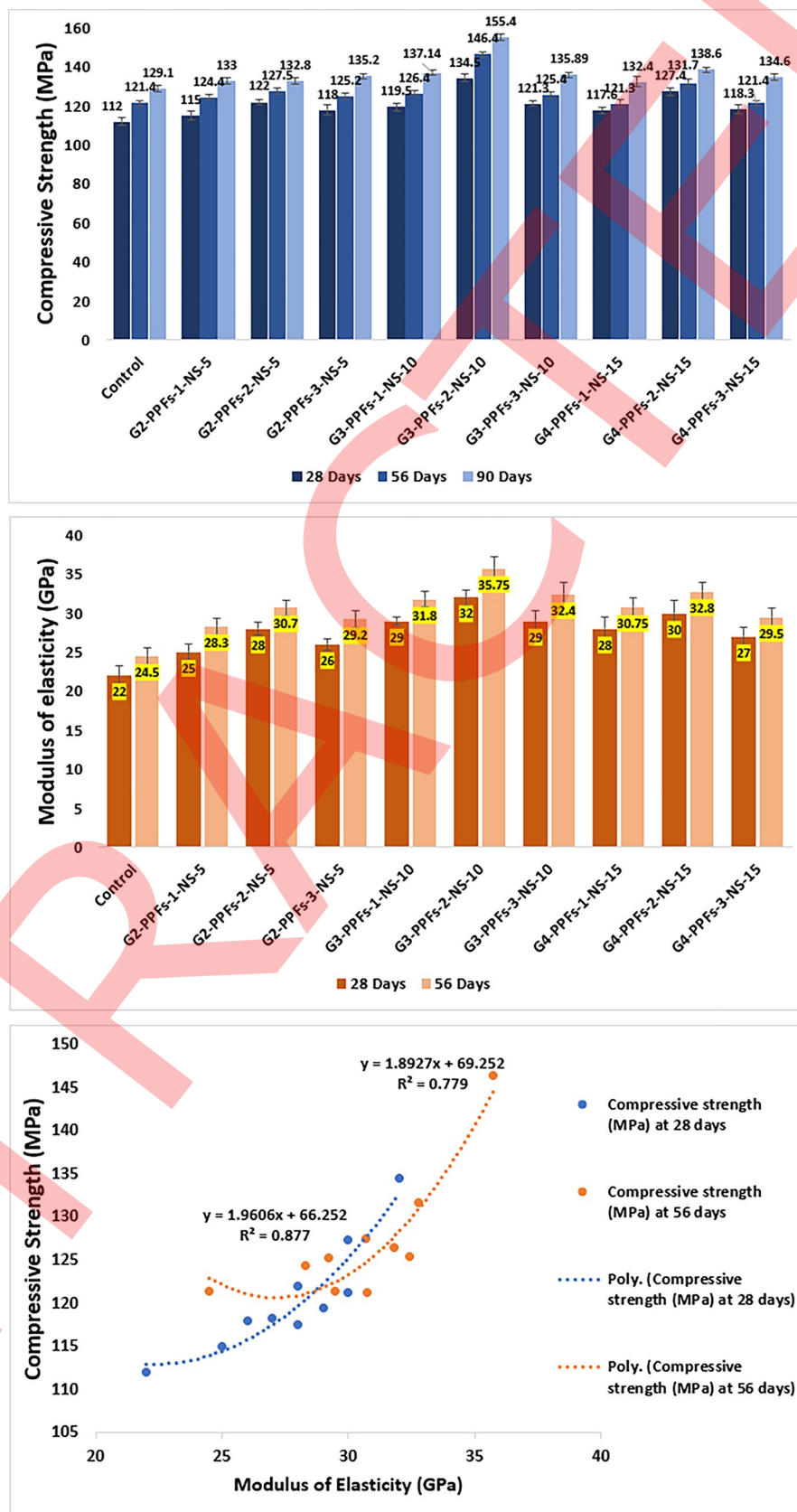


Fig 10. (a) Compressive strength of samples (MPa) at 28, 56, and 90 days. (b) Modulus of elasticity of UHS-GPC (GPa) at 28 and 56 days. (c) Co-relation for Compression strength and MOE at the curing of 28 and 56 days.

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strength and capacity in tensile strain. Because of the matrix's quality enhancement because of the micro-filling impact and pozzolanic behavior of nano-silica [32, 76], the inclusion of nano-silica could also improve the pullout capability of concrete structural members [77]. After the failure of concrete samples, the steel reinforcement was detached from the control samples, while the same steel reinforcement was still attached firmly to the concrete even after failing due to the presence of PPFs. The same observation was also noticed in the past study [78], in which the authors revealed a 15 to 25% enhancement in resistance against the pullout test because of the inclusion of 1 and 2% steel fibers. In another study, Harajli et al. [79] reported that samples with 2% fibers improved bond strength up to 60% compared to samples with no fibers. Furthermore, the bond strength was enhanced by 4 or 5 times with the inclusion of fibers in 3 and 4% [79]. Hence, it could be emphasized that the joint usage of nano-silica and PPFs improved the bond strength of concrete samples because of the improved adhesion of nano-silica and crack bridging impact of polypropylene fibers.

4.4 Splitting tensile strength (STS)

The effect of NS and fibers on STS is displayed in Fig 12(A). As with compressive strength performance, increasing the PPFs in samples improved the STS of UHS-GPC. When the number of PPFs was increased from 1% to 3%, the STS was enhanced from the control sample in every variant of the sample. When NS was added, the STS was improved at first but then reduced

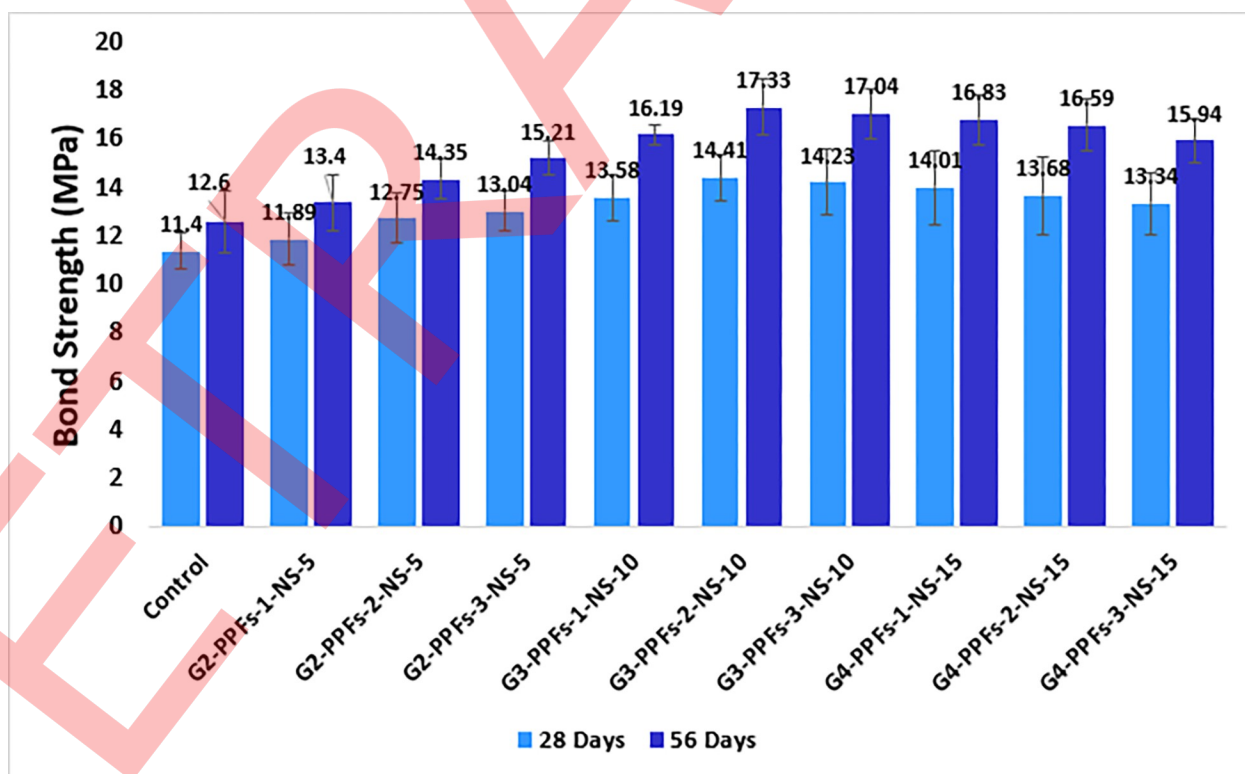


Fig 11. Bond strength of UHS-GPC at 28 and 56 days.

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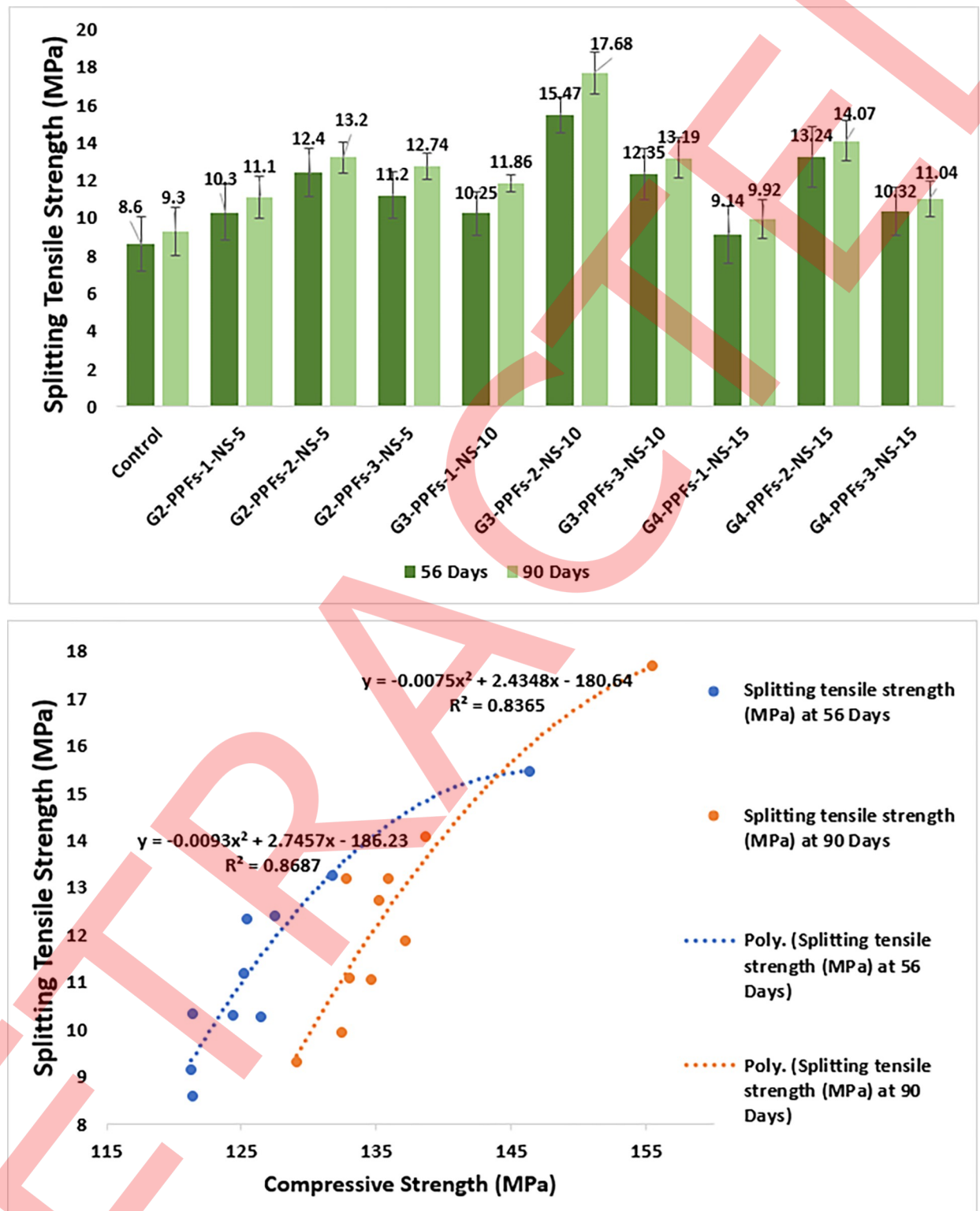


Fig 12. (a) Splitting tensile strength of UHS-GPC at 56 and 90 days. (b) Polynomial regression analysis to predict splitting tensile analysis employing compressive strength (MPa).

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when NS was added from 10% to 15%. The lower STS was 9.14 MPa at 56 days for 1% PPFs and 15% NS, and at 56 and 90 days, the highest splitting tensile strength was 15.4 MPa and 17.6 MPa at 10% NS and 2% PPFs which is 44.15% and 47.1% more splitting tensile strength than control sample at 56 and 90 days. When the NS is added by 5%, the properties of alkaline solutions change, which alters the formation of reaction products and the strength development. As per the previous research, the inclusion of silica fume and slag-based GPC slightly enhanced the compression strength but significantly improved the bonding strength between the fibers and matrix. It can be noted that with more than 10% nano-silica, the strength was reduced, which could be ascribed to surplus unreacted nano-silica, which leads to extreme self-hydration in the concrete mixture, leading to the development of cracking and so reduces the splitting tensile strength. Similar behavior of geopolymer concrete was reported in past studies [80, 81]. To estimate the strength and progress of a co-relation among entire strength numbers of compressive and splitting tensile strength, polynomial regression analysis is performed as depicted in Fig 10 to provide a polynomial relationship. The statistical analysis in Fig 12(B) permits estimating the values of splitting tensile strength for a particular specimen employing its compression strength values at 56 and 90 days. R^2 was noted to be near unity and estimated values with more than 80% constancy for split tensile strength, demonstrating the accuracy of the test outcomes.

4.5 Flexural strength

Fig 13 depicts the flexural strength of ultra-high-strength geopolymer concrete with different doses of PPFs and NS. The inclusion of PPFs enhances flexural strength. Flexural strength improved from 22.7 MPa to 25.43 MPa with 1 to 2% of PPFs at 90 days. More PPFs could

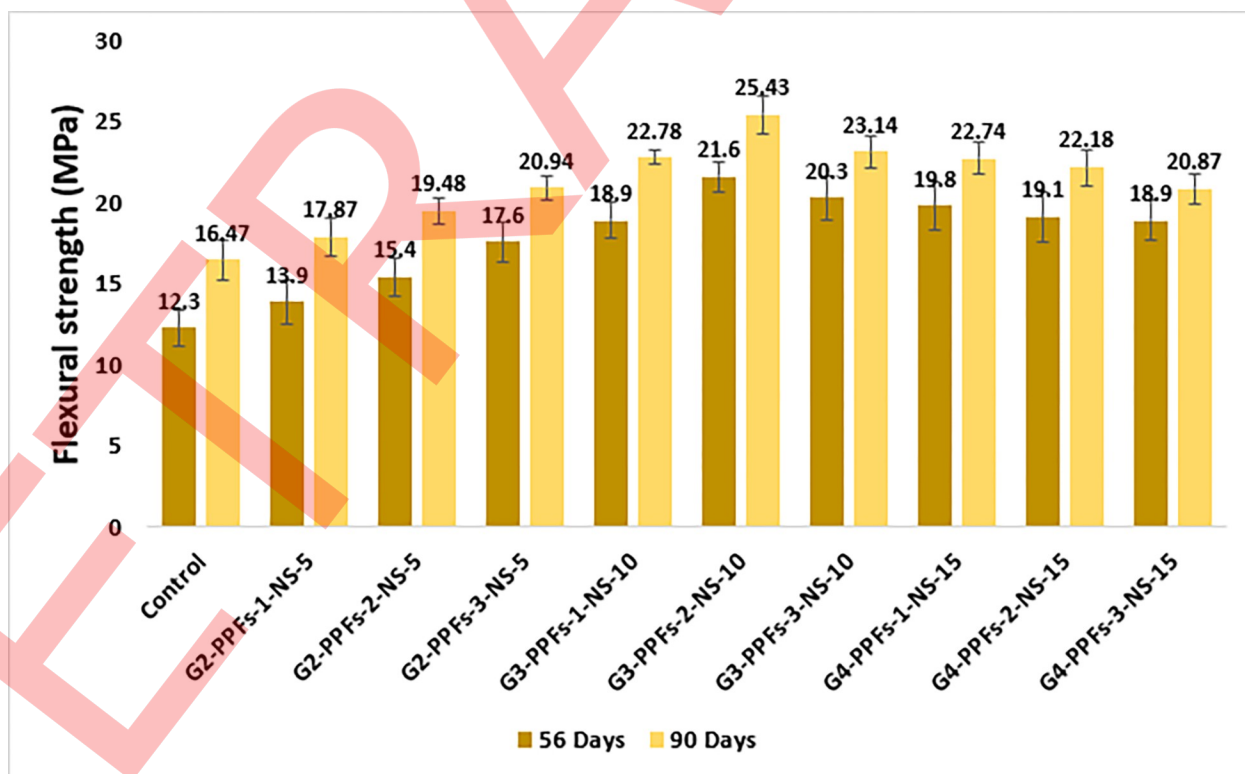


Fig 13. Flexural strength of UHS-GPC at 56 and 90 days.

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Fig 14. Concrete Sample after flexural test, (a) Control sample, (b) Modified Sample (G3-PPFs-2-NS-10).

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enhance the contact area between the polypropylene fibers and the concrete matrix, thus improving the matrix's capability to sustain the external load [31]. However, the amount of nano-silica also considerably influences the FS. The complex behavior of NS is due to a balance between the packed microstructure and the altered chemical activator [82]. At 90 days, the sample with 2% PPFs and 10% NS has a 36.52% higher flexural strength than other GPC samples. Fig 14A and 14B present the concrete samples after the flexural test. It could be noted from Fig 14(A), which is control concrete, that after the flexural test, it has large cracks down the middle portion of the beam and some minor cracks surrounding the beam. Fig 14(B) shows the modified sample (G3-PPFs-2-NS-10) in which there is an average crack in the mid portion of the concrete beam. Still, the crack is not propagated down the beam, and there are also no minor cracks around the beam which show the improvement of flexural strength in

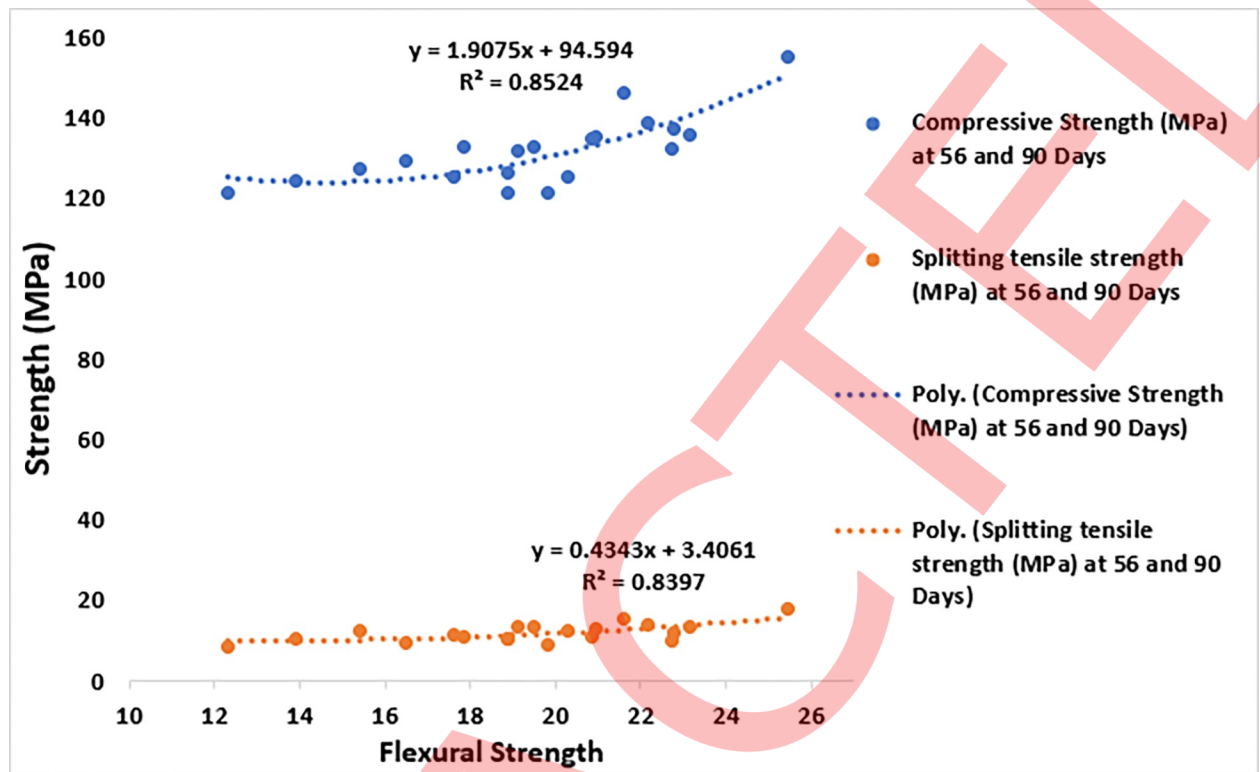


Fig 15. Polynomial regression analysis to predict flexural strength utilizing compressive and split tensile strength (MPa) at 56 and 90 days.

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the modified sample. The higher ultimate flexural strength could be because of the improved bonding strength amid PPFs and concrete matrix because of the inclusion of nano-silica [83, 84]. The test result shows that the joint utilization of PPFs and nano-silica attained excellent bending behavior.

To evaluate the flexural strength and develop a statistical co-relation among entire strength numbers of compressive and splitting tensile strength, linear regression analysis is carried out, as represented in Fig 15, to provide a statistical relationship. The statistical relationship in Fig 15 allows for assessing the flexural strength at 56 and 90 days by using the strength values of splitting tensile and compressive strength at 56 and 90 days. Values of R^2 were noted to be near unity and projected values with the consistency of more than 80% for flexural strength, which establishes the correctness of the test outcomes.

4.6 Load displacement test

Fig 16 depicts the load-displacement curve of UHS-GPC's specimens under bending load. At the start, a straight slope line was noticed up to the primary crack, and strain relaxation was noticed for every sample of UHS-GPC. The load-displacement curve of the control sample showed the same response as compared with the modified sample up to its first cracking, but when then polypropylene fibers were added. The joint impact of both nano-silica and PPFs was examined. It was noted that concrete's ductility is enhanced by adding nano-silica and PPFs up to an optimal dose of 10% nano-silica and 2% PPFs. This optimal mix displayed toughness, high residual strength, and lower load relaxations compared to samples without PPFs and nano-silica [27, 85]. This test showed that the polypropylene fibers and concrete matrix sustained the external load before the crack in the concrete sample. Adding more

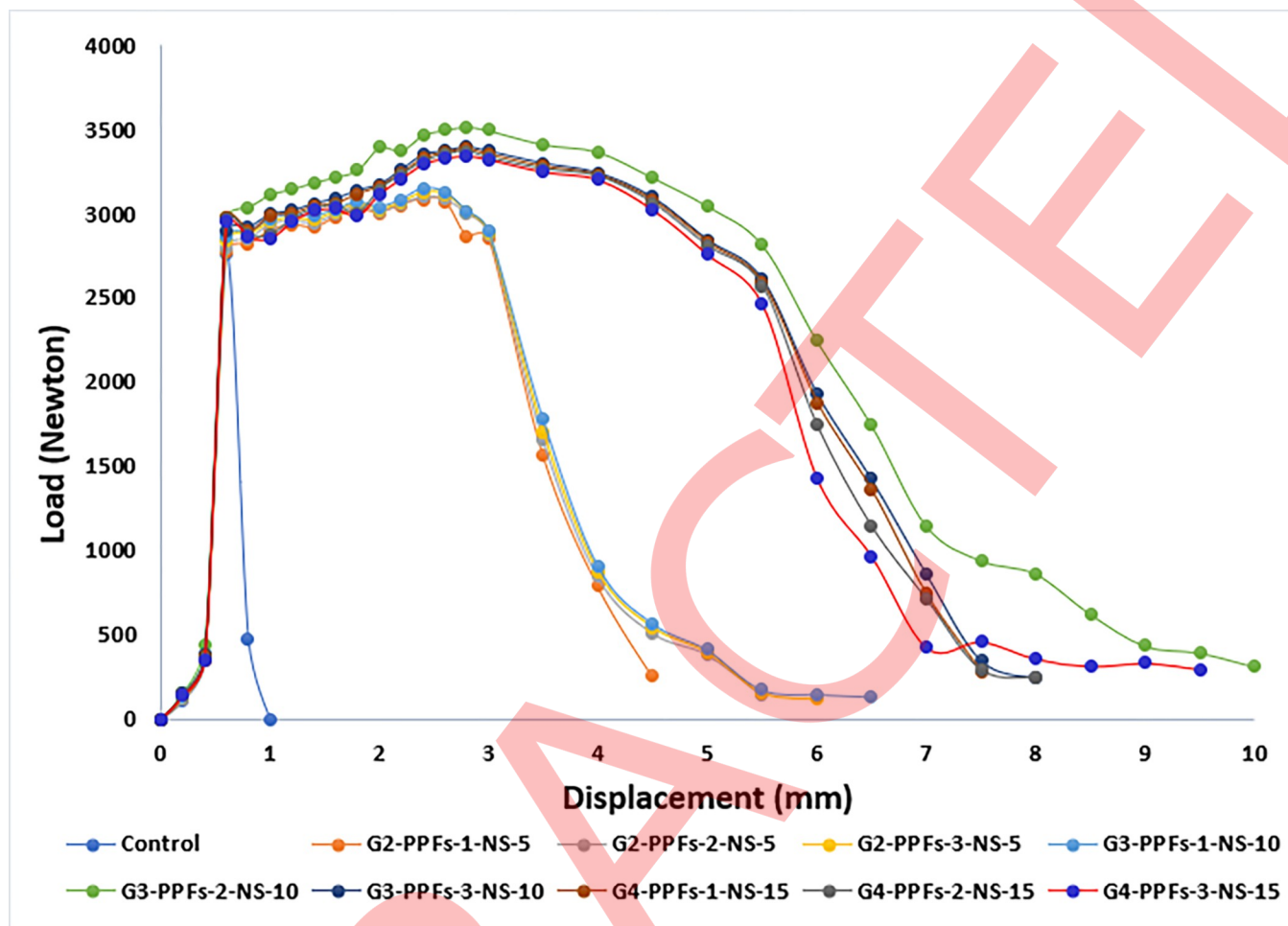


Fig 16. Load displacement curve for all samples.

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polypropylene fibers stiffens the sample, due to which, at 3% PPFs, the samples have lower ductility than samples with 2% PPFs. During the after-crack process, the cracked part is entirely tolerated by polypropylene fibers. Thus, the energy utilized during UHS-GPC failure enhances the proportion of PPFs, improving the ductility of ultra-high-strength geopolymer fiber-reinforced concrete.

4.7 Drying shrinkage performance of UHS-GPC

As discussed earlier, a drying shrinkage test was performed on all the concrete samples at 3, 7, 14, 21, 28, 56, and 90 days of age and then compared with the control concrete to assess the improvement in the performance of drying shrinkage. The test outcome after the 56 days was the base for determining the drying shrinkage because, after the 56 days, the strain values become stable. The test results of drying shrinkage are presented in Fig 17. When compared with the control sample at the age of 90 days, the drying shrinkage was decreased by 12.97% at 2% PPFs and 10% NS, which is very significant. This considerable reduction in micro-strains could be ascribed to the absence of moisture loss in modified samples. The reduced drying shrinkage of ultra-high-strength geopolymer concrete modified with PPFs and NS is credited to the existence of a less interconnected network of the capillary system in the concrete matrix.

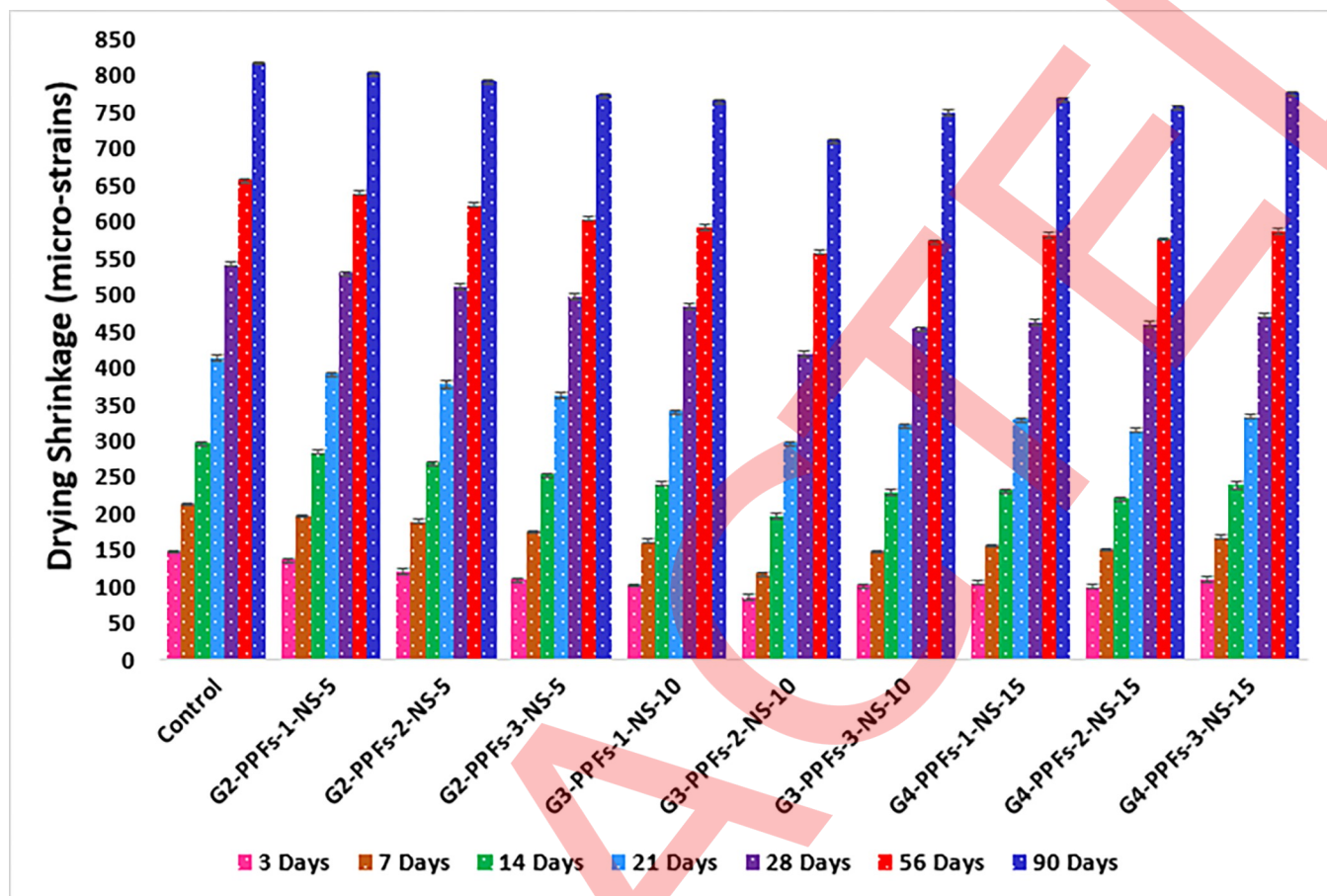


Fig 17. Drying shrinkage result of UHS-GPC.

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The drying shrinkage can also be impacted by the stiffness and porosity of aggregates used in the concrete. Micro-cracks are restricted by using polypropylene fibers, which confines the propagation of harmful agents into the concrete; PPFs also reduce the degree of cracks. At 3% PPFs and 15% NS, the drying shrinkage was observed to be increased by 7.05% at 90 days compared to the sample with 1% PPFs and 10% NS. This can be ascribed to nano-silica behaving like an activator in the hydration method, enhancing the degree of hydration and increasing drying shrinkage. The study's drying shrinkage results are consistent with the past literature [26, 86–88].

4.8 Fracture performance of ultra-high-strength geopolymer concrete

The fracture performance of concrete samples was evaluated utilizing Formula (C), which is provided in Fig 18(A). When the polypropylene fibers and nano-silica were added to the samples, it was noticed that the fracture energy was enhanced by raising the proportion of both polypropylene fibers and nano-silica at some specific level. The positive impact of polypropylene fibers was that the PPFs bridge the concrete cracks and help reduce them, averting the cracks' openings. Furthermore, nano-silica could develop bonding strength and improve the adhesion between the concrete's matrix and PPFs. The joint utilization of PPFs and nano-silica showed excellent fracture performance for the UHS-GPC samples. Another motive for the excellent fracture energy could be ascribed to improvement in the sample's compression

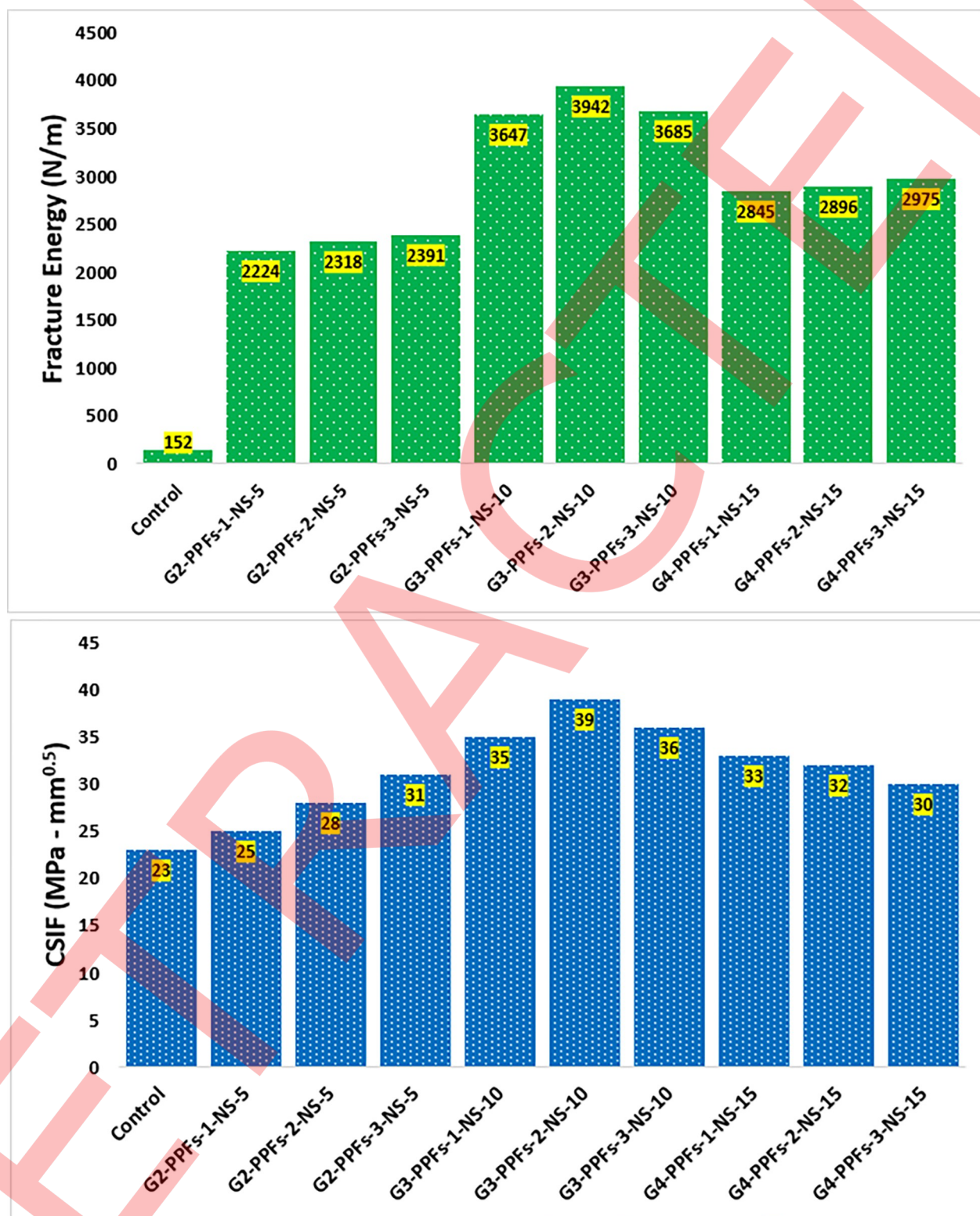


Fig 18. (a) Impact of PPFS and NS on UHS-GPC's fracture energy, (b) Impact of PPFS and NS on UHS-GPC's CSIF.

<https://doi.org/10.1371/journal.pone.0282435.g018>

strength having PPFs and nano-silica. In another study, the authors [89] studied the change of fracture energy in fly ash GPC specimens. They observed that the fracture energy enhanced as the concrete's compression strength improved. The critical stress intensity factor (CSIF) of UHS-GPC specimens was evaluated at 28 days of age utilizing the Formula (F), and the outcomes are provided in Fig 18(B). The critical stress intensity factor denotes the intensity of stress needed to spread the cracking. Similar outcomes of CSIF were attained for samples having various amounts of PPFs and nano-silica. When the percentage of polypropylene fibers and nano-silica in concrete samples were increased to 2% and 10%, respectively, it was observed that more stresses were needed to open the present cracking in the concrete sample. The improvement in CSIF could be ascribed to the high bonding because of the rise in the amount of nano-silica and PPFs up to optimal dose and the higher ability of PPFs to arrest cracks.

4.9 Performance of UHS-GPC against elevated temperature

Loss in weight and the residual compressive strength of all concrete samples were evaluated after their exposure to the specific elevated temperature. The samples were tested after 24 hours after their exposure to high heat. The results of the loss in weight of concrete samples after their exposure to high heat are presented in Fig 19(A). It could be supposed that using PPFs lowers weight loss relatively to high heat. Less loss in weight was observed in the sample with 2% PPFs and 10% NS due to the balanced amount of fibers. The highest loss in weight was observed at 750°C for all samples. At 750°C, the modified samples lost weight up to 43%, the geopolymer concrete was highly damaged after the high heat impact, and the concrete cracking became very wide. Some pieces of concrete were spalled from the rest of the sample, as displayed in Fig 19(B). With the temperature rise, dehydration gets higher in the concrete matrix, and all the moisture discharges by travelling to the concrete surface, which damages the concrete's microstructure and instigates a loss in the weight of concrete samples. The compressive strength of concrete samples after their exposure to elevated temperature is shown in Fig 20. It was observed that as the temperature was raised, the compressive strength of concrete samples reduced. The lowest compression strength observed at 750°C in modified samples

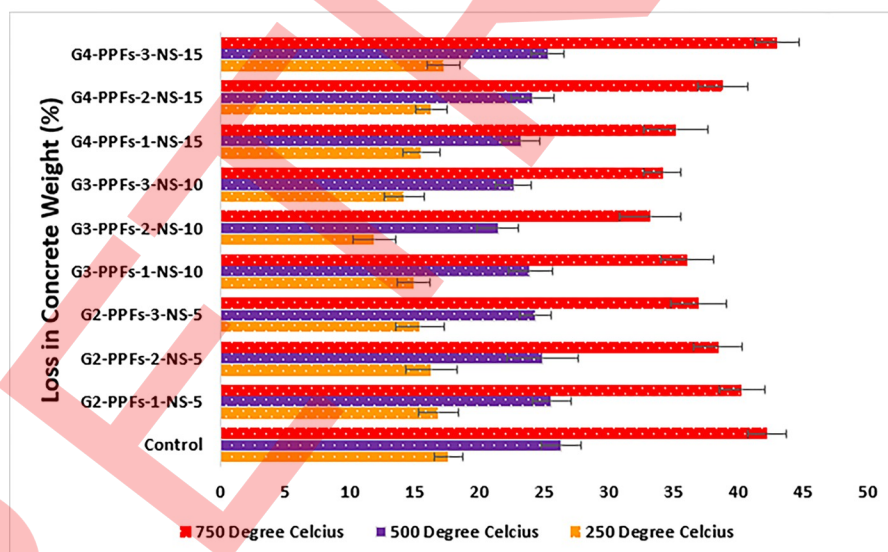


Fig 19. (a) Loss in UHS-GPC weight due to elevated temperature, (b) Concrete sample after exposure to elevated temperature.

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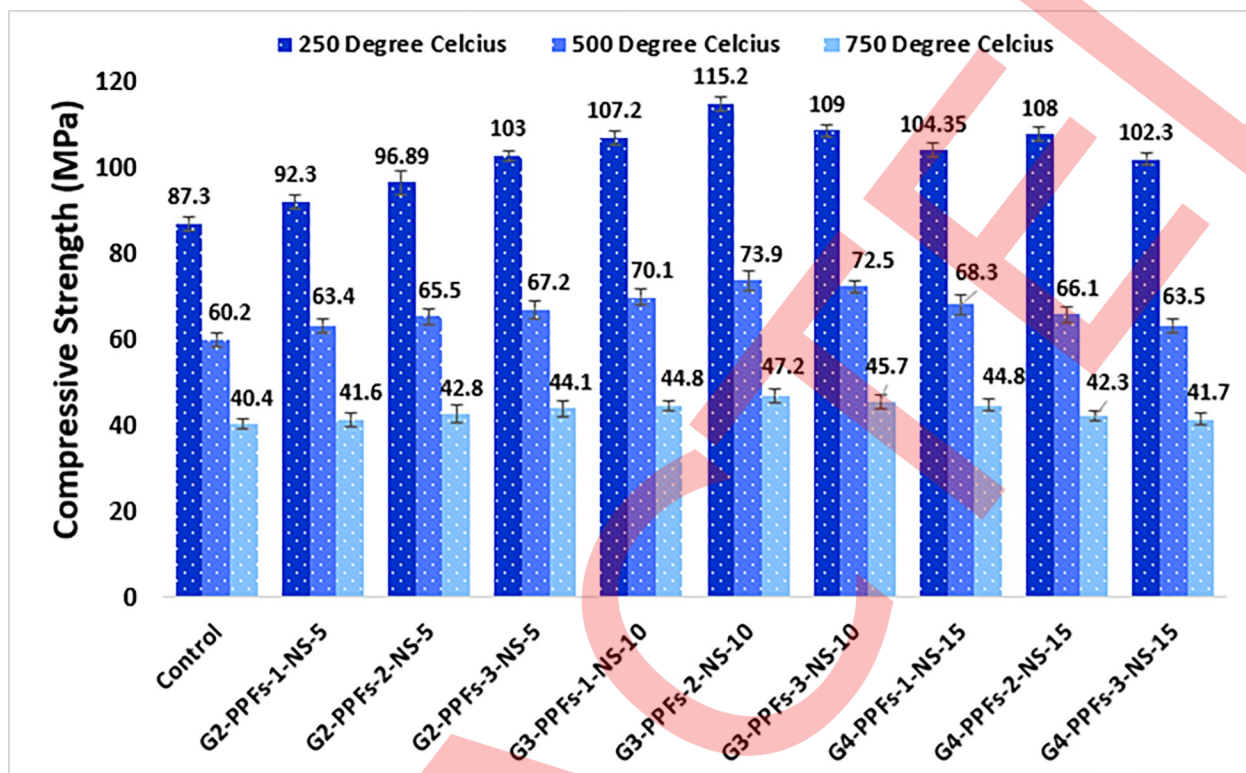


Fig 20. Compressive strength of UHS-GPC after elevated temperature.

<https://doi.org/10.1371/journal.pone.0282435.g020>

was 41.7 MPa, which was still slightly stronger than the control concrete at a similar temperature. This signifies the suitability of using PPFs and NS in UHS-GPC. The concrete samples exposed to 250°C had better resistance to heat than other temperatures. Heating the concrete causes the release of entrapped water or moisture, and this water or moisture has a considerable impact on the geopolymer concrete. It was mentioned in the research literature [90, 91] that water or moisture content within concrete specimens softens the binder gel or lowers the surface forces amid the gel particles, reducing the strength. Hence, removing water from the concrete mix by giving it heat up to 250°C helps attain good resistance against high heat [92]. Furthermore, increasing the temperature from 250°C to 500°C and 750°C leads to a significant loss in compressive strength. The significant reduction in compressive strength of UHS-GPC after 250°C could be due to the following reasons: (a) the development of micro-cracks in the interfacial transition zone amid the polypropylene fibers or aggregates and the concrete's matrix because of the difference in the coefficient of thermal between the components of UHS-GPC [93]; (b) decomposition of hydrates and phase-shift due to high heat which causes the strength reduction and existences of micro-cracking due to drying of samples [94].

5. Conclusions

The impact of PPFs and NS on the strength properties of UHS-GPC are considered and conferred in the present research. The subsequent conclusions are obtained from the current work.

1. The addition of PPFs decreased the slump value of freshly mixed UHS-GPC samples.

2. The maximum improvement of bond strength was noted to be 14.41 MPa and 17.3 MPa, which is the improvement of 26.47% and 37.58% at 28 and 56 days for the sample with 10% nano-silica and 2% PPFs.
3. When the amount of NS was more than 5%, i.e., at 10% NS, a firm bonding characteristic was observed among the PPFs and the matrix.
4. The highest splitting tensile strength was 15.4 MPa and 17.6 MPa at 10% NS and 2% PPFs which is 44.15% and 47.1% more splitting tensile strength than the control sample at 56 and 90 days.
5. The co-relation analysis showed the accuracy of test results as the values of the R square tended to be close to unity.
6. The inclusion of PPFs enhances flexural strength. The FS improved from 22.7 MPa to 25.43 MPa with 1 to 2% of PPFs at 90 days.
7. The load-displacement curve of the reference specimen displayed similar behavior as compared with the modified sample up to its first cracking. Still, when PPFs and NS were added, the load-displacement curve went up, and the sample with 2% PPFs and 10% NS had the highest load-carrying capacity against its corresponding load.
8. When compared with the control sample at the age of 90 days, the drying shrinkage was decreased by 12.97% at 2% PPFs and 10% NS, which is very significant.
9. When the PPFs and NS were added to the concrete, it was noticed that fracture energy was enhanced by increasing the polypropylene fibers and nano-silica.
10. When the percentage of PPFs and NS in concrete was raised to 2% and 10%, it was noted that more stresses were needed to open the present cracking in the concrete sample, which improved the critical stress intensity factor.
11. During the elevated temperature test, the concrete lost the minimum weight and compressive strength at 250°C and yielded the highest weight and compressive strength at 750°C.

Supporting information

S1 Table.
(XLSX)

Author Contributions

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Methodology: Fadi Althoei.

Resources: Saleh Alsulamy.

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Supervision: Rebeca Martínez-García.

Validation: Mohamed M. Arbili.

Visualization: Osama Zaid.

Writing – original draft: Osama Zaid.

Writing – review & editing: Saleh Alsulamy, Jesús de Prado-Gil, Mohamed M. Arbili.

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